Dr. HRISHIKESH CHANDRA GAUTAM

Contact Information

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Summary

Accomplished Air Quality Researcher with over five years of experience at EPIC-AQLI and CSTEP, specializing in urban air quality modeling, data analytics, and machine learning applications. Proficient in managing complex projects, with a proven track record in developing emission inventories, air quality action plans, and satellite data monitoring systems. Skilled in leveraging statistical analysis, data visualization, and health risk assessments to deliver actionable insights for policy and advocacy. Adept at stakeholder engagement, with a strong ability to simplify complex air quality data for diverse audiences. Proactive, collaborative, and committed to driving evidence-informed solutions for global sustainability, with a mission to enhance air quality data accessibility and impact.

Research Interests

Air quality management, Data analytics and visualization, Air Quality Index (AQI), Emission inventory development, Artificial Neural Network (ANN), Machine learning, Deep learning

Education

2012-2020: **Ph.D**
Indian Institute of Technology Madras, Chennai
Discipline: *Environmental engineering*

2010 - 2012: **M.Tech**
National Institute of Technology Karnataka, Surathkal
Discipline: *Environmental Engineering*

2005 - 2009: **B.Tech**
IMS Engineering College, Ghaziabad
(Uttar-Pradesh Technical University, Lucknow)
Discipline: *Biotechnology*

Professional experience

Feb, 2024 to present: **Data Specialist - Air Quality Life Index**
Energy Policy Institute at University of Chicago (EPIC)

- Spearheaded the execution of the Air Quality Life Index (AQLI) analysis and reporting pipeline, driving impactful insights on particulate pollution and its effects on life expectancy to inform global policy and advocacy. Key responsibilities include:
  - Managed an end-to-end data pipeline, from processing satellite-based PM2.5 data to calculating population-weighted concentrations, ensuring robust data synthesis and visualization for stakeholder engagement and impactful storytelling.
  - Performed time-series analysis on 25 years of GADM Level 2 data, uncovering critical trends in air quality and life expectancy across diverse regions, supporting strategic research and advocacy initiatives
  - Maintained a high-standard GitHub repository, enforcing clean code practices, seamless version control, and collaborative workflows

May, 2020 to Feb, 2024 : **Senior Associate**
Center for the study of science, technology and policy (CSTEP), Bangalore

- Developed Comprehensive Emission Inventories for 76 cities in India
  - Collected and synthesized secondary emission activity data from 76 cities.
  - Proactively engaged and coordinated with State Pollution Control Boards (SPCBs) officials to secure critical data, fostering strong stakeholder partnerships.
  - Leveraged advanced web-scraping tools to extract activity data from online resources, streamlining data acquisition for efficient analysis.
  - Conducted analysis of ground survey data, delivering actionable insights to inform policy and advocacy efforts.
- Created Air Pollution Emission Inventories for Six Cities in Jharkhand, India
  - Gathered and analysed secondary emission data across polluting sectors.
  - Performed scenario analysis and techno-economic assessments to develop evidence-based mitigation strategies, driving policy recommendations.
- Formulated State-Level Air Quality Plan for Punjab
  - Executed collection, preprocessing, and analysis of satellite-derived air quality data.
  - Validated and analyzed air quality data from low-cost sensors, ensuring accuracy and reliability for regional air quality assessments.
- Developed Advanced $PM_{2.5}$ Forecasting Models
  - Designed and implemented machine learning models (Random Forest, XGBoost, LSTM, SVR) to forecast $PM_{2.5}$ concentrations, enhancing predictive accuracy for air quality management.
- PAVITRA (air Pollution mAnagement and interVentIon Tool foR IndiA)
  - A collaboration between Indian Institute of Technology Bombay (IITB), CSTEP, University of California, Berkeley and the University of Washington (funded by Open Philanthropy)
  - Developing a reduced complexity model similar to InMAP for Indian cities
- Impact Assessment for DRIIV Project
  - A collaboration between Indian Institute of Technology Delhi (IITD), CSTEP and Air pollution action group (APAG)
  - Study for understanding the effect of abatement measures like pavement design on air quality at multiple locations in Delhi city using Low cost sensor (LCS) based monitoring

February, 2019 to June, 2019: **Senior Project Officer**
Indian Institute of Technology Madras, Chennai
Project Title: *Clean Air for Delhi Through Intervention, Mitigation and Engagement (CADTIME)*

- Project of Atmospheric Pollution and Human Health in an Indian Megacity (APHH) research programme funded by Natural Environment Research Council (NERC), UK and the Ministry of Earth Sciences (MoES), India
- Vehicular emission monitoring and analysis

- Development of statistical tools and predictive model for analysis of vehicular emissions in Delhi city

January, 2018 - May, 2018: **Senior Project Officer**
Indian Institute of Technology Madras, Chennai
Project Title: *Air Quality Management of Jawaharlal Nehru Port Trust (JNPT), Nhava Sheva, Mumbai*

- Setting up of air quality monitoring network at JNPT
- Execution of air quality monitoring plan for the area
- Monthly reporting of air quality status
- Development of management plan and application of mitigation measures

Software Skills

Programming: **Python** (**Data analysis**:Numpy, scipy, Pandas, **Data visualisation**: Matplotlib, Bokeh, Seaborn, plotly, dash, **Machine learning**: ScikitLearn, Tensorflow **Web scraping** Selenium, Requests, beautiful soup), **R** (ggplot2, Openair, R Shiny)

Softwares: MATLAB, LaTeX, SPSS, Origin, DBeaver

Operating System: Microsoft Windows, Linux (Ubuntu)

Research Experience

July, 2012 to April, 2020: *Ph.D. Scholar, Department of Civil Engineering, IIT Madras*
Urban air quality management using Neuro-Fuzzy modeling
Advisor: Dr. S M Shiva Nagendra, IIT Madras

- Understanding the statistical trend of criteria pollutants and their relationship with meteorological parameters
- Identification of statistical probability distribution of National Ambient Air Quality Standards (NAAQS) violations and prediction of their frequency using extreme value theory
- Development of an ANN based model to forecast air quality using pollutant and meteorological data in Chennai city
- Development of Fuzzy logic based AQI for estimation of air quality information and real-time health advisory.

August, 2011 - June, 2012: *M.Tech Project at National Institute of Technology Karnataka, Surathkal*
Assessment of groundwater quality in North and South Karanpura coalfields, Jharkhand
Advisor: Dr. Subhash Yaragal, National Institute of Technology Karnataka, Surathkal

January, 2009 - June, 2009: *B.Tech Project at IMS Engineering College, Ghaziabad*
Antimicrobial effects of common Indian spices
Advisor: Mrs. Shomini Parashar , IMS Engineering College, Ghaziabad

Certifications

How to Report Emissions under the Convention on Long-range Transboundary Air Pollution
UN CC:Learn Issued: June, 2023

Convention on Long-range Transboundary Air Pollution
UN CC:Learn Issued: June, 2023

Specialized module on cities and climate change
UN CC:Learn Issued: April, 2022

The Sustainable Development Goals – A global, transdisciplinary vision for the future
Coursera Issued: May, 2022

Introduction to ESG
Corporate Finance Instute (CFI) Issued: March, 2022

Induction training on National Clean Air Program (NCAP)
World bank Group Issued: February, 2022

Industrial Training

May, 2008 to July, 2008: *Central Research Institute (CRI), Kasauli*
Vaccine and Antiserum development

May, 2011 to July, 2011: *Campco chocolate factory, Puttur*
Wastewater treatment of Industrial effluent

Publications

Journal papers, Reports and Book chapters:

1. *Air Quality Life Index (AQLI). 2025. Annual Update AQLI report 2025*
2. *Air Quality Life Index (AQLI). 2024. Annual Update AQLI report 2024*
3. *CSTEP (2023) Air Pollution Emission Inventory for Six Cities in Jharkhand (CSTEP-RR-2023-5) https://www.cstep.in/publications-details.php?id=2354*
4. *Singh S., Agrawal P., P Kulkarni P., Gautam H.C., Kushwaha M., Vakacherla S. (2023)Multiple PM Low-Cost Sensors, Multiple Seasons' Data, and Multiple Calibration Models* Aerosol and Air Quality Research, 2023 *https://doi.org/10.4209/aaqr.220428*
5. *Kulkarni P., Vakacherla S., Upadhya A.R., Gautam H.C.(2022)Which model to choose? Performance comparison of statistical and machine learning models in predicting PM2.5 from high-resolution satellite Aerosol Optical Depth,* Atmos Env. Volume 282, 2022, 119164 https://doi.org/10.1016/j.atmosenv.2022.119164
6. Gautam H.C., Yadav V, Singh V. (2022) *IoT -enabled services for sustainable municipal solid waste management in India.* In book "IoT-based Smart Waste Management for Environmental Sustainability" *p.83-98. 1st* ed, CRC Press, Taylor Francis Group. *(ISBN:9781003184096)* *https://doi.org/10.1201/9781003184096-5*
7. Gautam H.C., Shiva Nagendra S.M. (2019) *"Study of meteorological impact on air quality in a humid tropical urban area,J Earth Syst Sci 128: 118* *https://doi.org/10.1007/s12040-019-1116-7*
8. Gautam H.C., Shiva Nagendra S.M. (2018) *Comparison of Fuzzy Synthetic Evaluation Techniques for Evaluation of Air Quality: A Case Study In:* Recent Developments and the New Direction in Soft-Computing Foundations and Applications. *Studies in Fuzziness and Soft Computing, vol 361. Springer, Cham* *https://doi.org/10.1007/978-3-319-75408-6_47*

Conference Papers:

1. *Agrawal, P., Srishti, S., Kulkarni, P., Gautam, H., Kushwaha, M., Vakacherla, S., and Singh, P. Correcting PM2.5 data from low-cost sensors using machine learning techniques, EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023, EGU23-3110, https://doi.org/10.5194/egusphere-egu23-3110, 2023*
2. *Singh, P., Banerjee, A., Kumar, U., Tiwari, A., Gautam, H., and Mishra, S. Emission Inventories for 76 Cities In India for Clean Air Strategies, EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023, EGU23-6701, https://doi.org/10.5194/egusphere-egu23-6701*
3. *Gautam H.C., Shiva Nagendra S.M. (2013) "Neuro-fuzzy applications in urban air quality management", in International Conference on Advanced Engineering Optimization Through Intelligent Techniques held in July 01-03, 2013 at S.V. National Institute of Technology, Surat – 395 007, Gujarat, India*
4. *Gautam H.C., Shiva Nagendra S.M. (2014) " NO_2 forecasting using Artificial Neural Network for development of early warning system for Manali Industrial area in Chennai, India", in Better Air Quality conference 2014, Colombo, Srilanka*
5. *Gautam H.C., Shiva Nagendra S.M. (2015) "Development of Fuzzy inference system for assessment of air quality in an urban area, presented as poster in Seventh Workshop on Big Data Benchmarking, 2015, New Delhi*

Extra curricular activities

- ***Member of organising committee*** *in* ***India Clean Air Summit*** *Conference organised by CSTEP Bengaluru, 2020 - 2023*
- ***Member of organising committee*** *in* ***IICAQM*** *Conference held in IIT Madras, 2016*
- ***Won*** 1^{st} ***Prize*** *in* ***Back to Science Quiz*** *competition held on Research Scholars day, IIT Madras, 2015*
- ***Organising Secretary*** *of* ***"Water Conservation and Management"(WCM-2011)*** *Conference held on 22 March 2011 (World water day) at NITK Surathkal*
- ***Won*** 1^{st} ***Prize*** *in quiz competition in* ***Inter-Institutional 'BIOFEST-2009'*** *at I.M.S Engineering College, Ghaziabad*

References

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